Highlights

A Voxel-Level Characterization of Decoder Computation in 3D Medical Image Segmentation

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- Characterizes the spatial contribution of additional decoder computation in 3D segmentation.
- Establishes the distinction between correctness activity (gross changes) and net gain (directional benefit).
- Demonstrates boundary-localized bidirectional correctness changes and highly concentrated positive gain ($K_{80} \leq 5\%$).
- Demonstrates predictive uncertainty signals activity but provides weak directional discrimination.
- Evaluates retrospective output-space hybridization to recover macro Dice gaps without over-extrapolating runtime speedups.

A Voxel-Level Characterization of Decoder Computation in 3D Medical Image Segmentation

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Abstract

Modern 3D medical image segmentation networks with vision transformers and large-kernel CNNs increasingly incorporate heavy multi-resolution decoders, assuming that dense decoding is universally necessary across the volumetric grid. Conversely, efficient alternatives simplify these decoding pathways based on macroscopic metrics. However, the locations, spatial conditions, and magnitudes to which extra decoder computation genuinely improves segmentation correctness remain uncharacterized. We present a systematic characterization of decoder computation in 3D medical image segmentation. We evaluate end-to-end full–efficient decoder pairs across three representative backbones on BTCV CT, examine cross-dataset consistency using 3D UX-Net across BTCV CT, MSD01 BrainTumour MRI, and MSD08 HepaticVessel CT, and perform controlled factor attribution using a frozen shared-encoder 2×2 factorial design on 3D UX-Net. Our characterization separates two quantities that aggregate metrics conflate: *correctness activity*—where the decoder changes a voxel's correctness—and *net gain*, the signed balance of corrections and degradations. The two diverge sharply. Across every backbone and dataset, added decoder capacity is highly active yet nearly net-neutral: large bidirectional activity, but a net shift statistically indistinguishable from zero. The residual gain is concentrated—an oracle captures $\geq 80\%$ of it within the top 5% of blocks ($K_{80} \leq 5\%$)—and confined to a thin anatomical-boundary band—the margins that govern surgical planning and radiotherapy targeting. Crucially, uncertainty predicts *where* predictions change (location AUROC 0.865–0.913) far better than *whether* they improve (direction AUROC 0.520–0.591); routing a 20% budget to the full decoder then recovers the macro-Dice gap wherever one exists. Prediction difficulty and computational utility are thus related but distinct. Clinically, this clarifies when efficient decoders can be trusted for latency-critical intraoperative use and where computation must be spent to protect boundary-sensitive decisions, bounding what future adaptive decoders can achieve.

Keywords: 3D Medical Image Segmentation, Efficient Deep Learning, Uncertainty Quantification, Decoder Computation, Model Characterization

1. Introduction

Precise 3D medical image segmentation is essential for clinical applications such as image-guided surgical planning, volumetric tumor monitoring, and targeted radiation therapy. In deep learning, encoder-decoder networks based on the U-Net paradigm [1, 2] remain the foundational standard. Recent research has primarily focused on scaling encoder expressiveness through vision transformers, large convolutional kernels, and self-supervised pre-training [3, 4**?** , 5]. In parallel, modern 3D architectures retain heavy multi-resolution decoders equipped with dense upsampling stages, complex skip-fusion connections, and spatial refinement blocks. Because 3D volumetric scans scale cubically in resolution ($O(N^3)$), these multi-scale decoders routinely account for over 40% of total inference FLOPs, posing severe memory and latency bottlenecks for real-time intraoperative deployment. This prevailing architectural trajectory implicitly assumes that dense, high-capacity decoding is universally necessary across every voxel in the 3D volume.

Conversely, emerging lightweight models attempt to mitigate this computational burden by simplifying decoding pathways, such as EffiDec3D [6] and EfficientMedNeXt [7]. Yet both the heavy and the lightweight camps are judged on the same footing: standard evaluation paradigms assess decoder designs almost exclusively through aggregate metrics such as the global Dice similarity coefficient or Hausdorff distance. Herein lies an information gap. A single volume-averaged scalar collapses millions of independent voxel-level decisions into one number, discarding precisely the spatial micro-dynamics it is meant to explain—which voxels the extra computation flips, in which direction, and under what anatomical conditions. A decoder that rewrites thousands of voxels near anatomical boundaries and one that leaves the volume nearly untouched can report an identical Dice score. This gap is not merely academic: the boundary voxels that aggregate metrics dilute are exactly the ones that determine clinical outcomes—tumor margins that dictate resection extent, vessel walls that constrain catheter navigation, and organ interfaces that bound radiation dose. A characterization that resolves *where* and *whether* decoder computation helps therefore speaks directly to whether a lighter, faster model can be trusted at the anatomical sites where errors carry the greatest

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clinical cost. Consequently, a fundamental scientific question remains unexamined: **How does extra decoder computation deliver spatially-varying performance gains for volumetric medical segmentation?**

To address this question, we systematically characterize decoder computation in 3D medical image segmentation models. Rather than proposing a new network architecture, we treat computation characterization as a distinct empirical approach. The main contributions of this study are:

1. We introduce a voxel-level framework that separates decoder correctness activity from directional net gain.
2. We characterize decoder contribution in terms of spatial heterogeneity, positive-gain concentration, and inference-time predictability.
3. We provide controlled evidence across architectures, datasets, and decoder interventions, and establish retrospective opportunity bounds for future utility-aware adaptive decoding.

The remainder of this paper is organized as follows: Section 2 reviews related literature and highlights key gaps. Section 3 details our characterization framework and metrics. Section 4 describes our experimental setup and evaluation protocols. Section 5 presents our observational findings, Section 6 discusses underlying mechanisms and limitations, and Section 7 concludes the paper.

2. Related Work

2.1. Decoder Design in 3D Medical Image Segmentation

The encoder-decoder architecture introduced by U-Net [1, 2] and V-Net [8] remains the central design pattern for 3D medical image segmentation. Frameworks such as nnU-Net [9] demonstrated the effectiveness of standardized training protocols combined with deep decoding pathways. Subsequent developments incorporated Transformer backbones and large-kernel convolutions to capture long-range contextual dependencies (e.g., TransUNet [10], UNETR [3], Swin UNETR [4], 3D UX-Net [**?**], and MedNeXt [5]). These modern architectures emphasize expressive feature extraction while maintaining dense multi-scale decoders. However, these designs are evaluated primarily through final segmentation accuracy and overall parameter count; how decoder capacity operates spatially across volume regions remains unexamined.

2.2. Efficient Segmentation Architecture Design

To mitigate the high computational cost of 3D volumetric convolutions, recent works have explored lightweight decoder designs. EffiDec3D [6] introduced systematic decoder pruning by reducing channel dimensions and eliminating high-resolution decoding stages, establishing trade-offs between parameter scale and Dice performance. Similarly, EfficientMedNeXt [7] simplified decoding stages before introducing residual blocks. Other lightweight models achieve efficiency through hybrid transformer-convolution encoders, attention pruning, or custom lightweight modules (e.g., SegFormer [11], UNETR++ [12], Slim UNETR [13]). Nevertheless, efficient architecture studies infer redundancy indirectly from global performance curves, leaving unexamined whether the eliminated computation was spatially uniform or concentrated in specific anatomical sub-regions.

2.3. Model Characterization and Uncertainty Quantification

Recent studies have explored adaptive inference for volumetric processing [14, 15]. Epistemic and aleatoric uncertainty metrics have been widely applied to quantify model confidence and detect out-of-distribution inputs [16, 17, 18]. Existing uncertainty-aware quality assessment methods estimate whether a current segmentation is likely to be accurate. In contrast, our study asks a counterfactual computational question: whether allocating additional decoder computation is likely to improve or degrade that segmentation. Uncertainty-driven dynamic execution in 3D medical segmentation is hampered by a critical unexamined gap: it remains unclear whether predictive uncertainty can locate regions where additional decoder computation alters correctness, and whether it can further distinguish beneficial corrections from harmful degradations. Our framework explicitly addresses these questions by separating correctness activity from directional net gain.

3. Methods: A Characterization Framework

To understand the spatial and functional contribution of decoder computation, we establish a controlled observation framework (Figure 1). This section outlines the controlled decoder interventions, voxel-level correctness framework, rate definitions, spatial block partitioning, and characterization metrics.

3.1. Controlled Decoder Interventions

We analyze decoder capacity by isolating two generic architectural factors that govern volumetric decoding computation:

1. **Channel Capacity ($C_{\mathbf{dec}}$)**: The feature channel dimension across upsampling layers, controlling feature representation width ($C_{\text{full}} \rightarrow C_{\text{reduced}}$).
2. **High-Resolution Decoder Stages (RF)**: The resolution scaling factor, controlling whether high-resolution upsampling stages (e.g., full resolution 1×) are retained (RF_1) or omitted (RF_2).

Crossing these two experimental factors forms a 2×2 factorial intervention design (Full E_0, Channel-reduced, Stage-reduced, and Combined-efficient E_1). The specific channel-reduction ratios and stage-removal configurations follow the architecture-preserving decoder simplification strategy introduced by EffiDec3D [6]. Interventions occur strictly within the decoder pathway (upsampling, skip fusion, and refinement) while keeping the feature encoder architecture untouched.

Choice of Intervention Template: We adopt the reduction principles of EffiDec3D as a controlled intervention template rather than treating EffiDec3D itself as the primary object of evaluation. Its decoder-localized and architecture-preserving

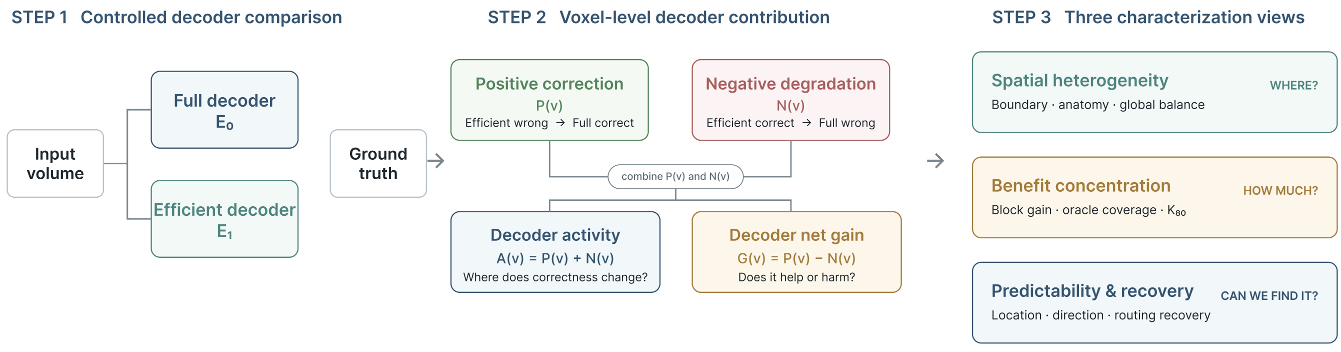


Figure 1: Overview of the decoder computation characterization framework across Step 1 (Controlled Interventions), Step 2 (Voxel Correctness Activity $A_{\text{corr}}(v)$ and Net Gain $G(v)$), and Step 3 (Three Characterization Axes).

modifications provide matched configurations in which channel capacity and high-resolution stage retention can be independently controlled. This enables spatial characterization of decoder contribution while minimizing confounding changes to the encoder family, prediction task, or training objective.

3.2. Voxel-Level Correctness Activity Framework

Let v denote a voxel within a 3D medical image volume V, with ground truth label $y(v)$, full decoder prediction $\hat{y}_f(v)$, and efficient decoder prediction $\hat{y}_e(v)$. Comparing predictions between matched E_0 and E_1 decoders yields two correctness-changing outcomes per voxel (Figure 2; see Supplementary Figures S1 and S2 for 3D UX-Net and MedNeXt-M-K3 visualizations):

- **Positive Correction** ($P(v)$): $P(v) = \mathbf{1}[\hat{y}_e(v) \neq y(v) \wedge \hat{y}_f(v) = y(v)]$, where the full decoder corrects an efficient baseline error.
- **Negative Degradation** ($N(v)$): $N(v) = \mathbf{1}[\hat{y}_e(v) = y(v) \wedge \hat{y}_f(v) \neq y(v)]$, where the full decoder alters a previously correct efficient prediction.

Defining voxel correctness indicators $C_f(v) = \mathbf{1}[\hat{y}_f(v) = y(v)]$ and $C_e(v) = \mathbf{1}[\hat{y}_e(v) = y(v)]$, we formalize two fundamental voxel indicators:

- **Decoder Net Gain** ($G(v) = C_f(v) - C_e(v) = P(v) - N(v)$): Measures the net directional balance ($G(v) = +1$ for correction, -1 for degradation, 0 for unchanged).
- **Correctness Activity** ($A_{\text{corr}}(v) = P(v) + N(v) = |G(v)|$): Measures where additional decoder capacity alters prediction correctness (gross activity).

3.3. Subject-Level Rate Formulations and Evaluated Domains

To aggregate voxel-level indicators into global rates for subject i, we define the subject-level positive correction rate $R_+^{(i)}$, negative degradation rate $R_-^{(i)}$, gross correctness activity rate $R_{\text{act}}^{(i)}$, and net gain rate $R_{\text{net}}^{(i)}$ over an evaluation domain Ω_i:

$$R_+^{(i)} = \frac{1}{|\Omega_i|} \sum_{v \in \Omega_i} P_i(v), \quad R_-^{(i)} = \frac{1}{|\Omega_i|} \sum_{v \in \Omega_i} N_i(v) \tag{1}$$

$$R_{\text{act}}^{(i)} = R_+^{(i)} + R_-^{(i)}, \quad R_{\text{net}}^{(i)} = R_+^{(i)} - R_-^{(i)} \tag{2}$$

We evaluate rates across two explicit spatial domains Ω_i:

1. **All-Volume Domain (Ω_{all})**: All voxels within the 3D bounding volume grid V_i, denoted by $R_{\text{act}}^{\text{all}}$ and $R_{\text{net}}^{\text{all}}$.
2. **Foreground-Union Domain (Ω_{fg})**: The subset of voxels belonging to ground-truth or predicted foreground structures: $\Omega_{\text{fg}} = \{v \in V_i \mid y_i(v) > 0 \vee \hat{y}_{e,i}(v) > 0 \vee \hat{y}_{f,i}(v) > 0\}$, denoted by $R_{\text{act}}^{\text{fg}}$ and $R_{\text{net}}^{\text{fg}}$.

Global metrics are reported as subject-level population means with 95% confidence intervals computed via 1000-resample paired bootstrapping across subjects.

3.4. Characterization Metrics and Spatial Partitioning

The voxel indicators of Section 3.2 and the subject-level rates of Section 3.3 quantify *how much* the decoder changes and *in which net direction*. They do not yet say *where* those changes land, *how concentrated* the beneficial ones are, or *whether* either property can be anticipated before the full decoder is ever run. This section develops the measures that answer these questions. We organize them along the three characterization axes summarized in Table 1: the **spatial heterogeneity** of correctness change relative to anatomy, the **concentration** of positive net gain across space, and the inference-time **predictability** of both the location and the direction of change.

All three axes are computed on a shared spatial substrate that makes measurements comparable across datasets with heterogeneous voxel spacings: a physical ground-truth boundary-distance stratification (Section 3.4.1) and a physically normalized block partition (Section 3.4.2). We then define the concentration measure K_{80} (Section 3.4.3) and the uncertainty-based predictability scores (Section 3.4.4). Finally, two analyses complement the axes and connect them back to practice: a retrospective output-space hybridization (Section 3.4.5) that upper-bounds the macro-metric benefit recoverable by localized decoding, and a factorial decomposition (Section 3.4.6) that attributes the observed effects to the two decoder factors of Section 3.1. Unless stated otherwise, block-level metrics are computed within each subject and summarized across subjects, with 95% confidence intervals from subject-level clustered bootstrap resampling.

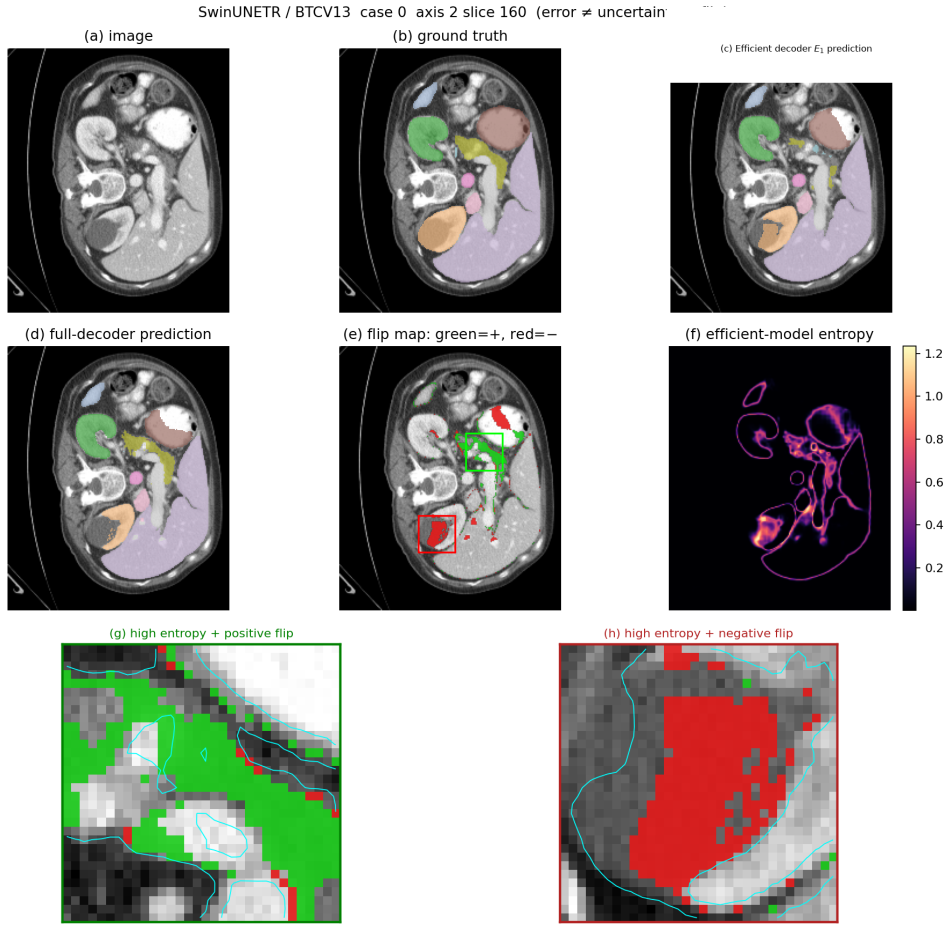


Figure 2: Qualitative illustration of correctness activity and directional gain for SwinUNETR on BTCV CT dataset under Full (E_0) vs. Efficient (E_1) decoders (see Supplementary Figures S1 and S2 for 3D UX-Net and MedNeXt-M-K3).

Table 1: Overview of characterization objectives and evaluation measures.

Analysis Objective	**Evaluation Measure**	**Question Answered**
Gross spatial effect	Correctness Activity $A_{\text{corr}}(v)$	Where does additional decoder computation change prediction correctness?
Directional effect	Net Gain $G(v)$	Where does the decoder improve correctness overall?
Benefit concentration	Oracle K_{80}	How spatially concentrated is positive net gain?
Activity predictability	Location AUROC	Can cheap signals detect where correctness changes occur?
Gain predictability	Direction AUROC	Can cheap signals distinguish beneficial from harmful changes?
Retrospective recovery	Retrospective Hybrid $R_{\text{rec}}(k)$	How much of the full decoder's Dice gap is recovered by output hybridization?
Multi-sample uncertainty	TTA Mutual Info $I(v)$	Does multi-view epistemic disagreement resolve direction predictability?

3.4.1. *Physical Ground-Truth Anatomical Boundary Distance*

The first axis asks how correctness change distributes relative to anatomy, and specifically whether it clusters at structure margins rather than in homogeneous interiors. Answering this requires a distance reference that is both anatomically meaningful and free of circularity. We therefore anchor all distance analysis strictly to the **ground-truth anatomical boundary**: measuring distance from a model's own predicted boundary would be circular, because that boundary is itself an output of the decoder under study and would shift with the very intervention we are characterizing. We further measure distance in physical millimeters rather than voxels. A fixed two-voxel neighborhood spans different physical extents across datasets with anisotropic spacing, so voxel-indexed bands would not describe a comparable anatomical margin from one modality to another. Incorporating the dataset voxel spacings $\mathbf{D} = \text{diag}(s_x, s_y, s_z)$, the physical boundary distance $d_{GT}(v)$ is

$$d_{GT}(v) = \min_{u \in \partial S_{GT}} \|\mathbf{D}(x_v - x_u)\|_2, \tag{3}$$

where ∂S_{GT} is the boundary voxel set of the ground-truth annotation $\bigcup_c (y(v) = c)$. We stratify correctness activity $A_{\text{corr}}(v)$ and net gain $G(v)$ across the physical distance bands 0–2, 2–4, 4–8, and $>$ 8 mm. The innermost band captures the immediate partial-volume interface where segmentation ambiguity concentrates; the outermost band captures the homogeneous organ interior, where a correctness change would indicate a qualitatively different, non-boundary effect. Reporting activity and net gain separately per band lets us test not only *whether* the decoder is active near boundaries but whether that activity translates into a directional benefit there.

3.4.2. Physically Normalized Spatial Block Partition

Individual correction and degradation events are sparse and spatially clustered, which makes the single voxel a noisy unit for the concentration and predictability analyses that follow. We therefore aggregate voxels into contiguous cubic blocks and treat the block as the unit of analysis. To keep a block anatomically comparable across datasets, volumes are resampled to dataset-specific target spacings (BTCV: $1.5 \times 1.5 \times 2.0$ mm^3; MSD01: $1.0\times1.0\times1.0$ mm^3; MSD08: $0.8\times0.8\times1.5$ mm^3) and block dimensions are chosen so that every block covers approximately the same physical volume of $24 \times 24 \times 24$ mm (BTCV: $16 \times 16 \times 12$ voxels; MSD01: $24 \times 24 \times 24$ voxels; MSD08: $30 \times 30 \times 16$ voxels). A block b_r is *active* if it contains any correctness change,

$$A_{\text{corr}}(r) = \mathbf{1}\Big[\textstyle\sum_{v\in b_r} A_{\text{corr}}(v) > 0\Big], \tag{4}$$

and its *directional* label is the sign of its aggregate net gain $G(r) = \sum_{v\in b_r} G(v)$. Direction analysis is restricted to blocks with $G(r) \neq 0$: a zero-gain block has no defined direction, and including such blocks would trivially inflate the majority class and obscure the actual difficulty of separating beneficial from harmful change. This active-versus-directional distinction is exactly what separates the location and direction predictability scores defined below.

3.4.3. Positive-Gain Concentration (K_{80})

A decoder can be net-neutral over the volume yet still harbor a small pocket of genuine, spatially concentrated benefit; the aggregate rates of Section 3.3 cannot distinguish that case from diffuse, uniformly distributed gain. The concentration axis quantifies this directly. Sorting blocks by descending net gain $G(r)$, the oracle coverage attained by the top $k\%$ of blocks is

$$\text{Coverage}(k) = \frac{\sum_{r=1}^{\lfloor (k/100)\cdot|B|\rfloor} \max(G(r), 0)}{\sum_{r=1}^{|B|} \max(G(r), 0)}, \tag{5}$$

which accumulates only positive contributions so that degradations do not mask the recoverable gain. We report K_{80}, the minimum percentage of blocks an oracle selector must retain to reach 80% coverage [19]. A small K_{80} means almost all of the decoder's useful work could in principle be reproduced by touching a tiny fraction of space, which is the necessary precondition for any budgeted adaptive decoder; a large K_{80} would imply the benefit is too diffuse to route around.

3.4.4. Uncertainty Signals and Predictability Scores

Concentration establishes that a small region *could* carry the benefit; predictability asks whether a cheap, single-pass signal available at inference time can actually *find* that region without running the full decoder. We use the predictive Shannon entropy of the efficient baseline,

$$H(v) = -\sum_{c=1}^{C} p_c(v) \log p_c(v), \tag{6}$$

as the canonical uncertainty signal [20, 16, 21], and reduce it to a block score $H(r)$ by volume averaging. To test whether multi-sample epistemic disagreement carries information that single-pass entropy misses, we additionally compute a Test-Time Augmentation Mutual Information (TTA-MI) score $I(r)$ over $K = 8$ spatial view transformations:

$$H(r) = \frac{1}{|b_r|} \sum_{v\in b_r} H(v), \quad I(r) = \frac{1}{|b_r|} \sum_{v\in b_r} I(v), \tag{7}$$

where $I(v) = H(\bar{p}(v)) - \frac{1}{K}\sum_{k=1}^{K} H(p_k(v))$; every augmented probability map is inverse-warped to the original image coordinate system before the ensemble mean and mutual information are formed. We then separate two questions that the literature routinely conflates. **Location predictability** (AUROC/AUPRC of $H(r)$ predicting $A_{\text{corr}}(r) > 0$) measures whether uncertainty finds *where* correctness changes. **Direction predictability** (AUROC/AUPRC of $H(r)$ predicting $G(r) > 0$ on non-zero-gain blocks) [22] measures whether it distinguishes *beneficial* change from *harmful* change. To expose *why* the two can diverge, we also report the **conditional density overlap** between $p(H \mid P(v) = 1)$ and $p(H \mid N(v) = 1)$: corrections and degradations that share the same entropy distribution are, by construction, directionally indistinguishable from entropy alone.

3.4.5. Retrospective Output-Space Hybridization (R_{rec})

The preceding axes describe the spatial structure of decoder benefit but do not, by themselves, establish that acting on that structure would recover the macro-level metric that practitioners optimize. This analysis closes that loop retrospectively. Given a selected block set S_k, we form a hybrid prediction that substitutes full-decoder outputs inside the selection while keeping the efficient baseline everywhere else:

$$\hat{y}_k(v) = \begin{cases} \hat{y}_f(v), & \text{if } v \in S_k \\ \hat{y}_e(v), & \text{otherwise.} \end{cases} \tag{8}$$

When the full decoder holds a positive macro-Dice advantage, $\text{Dice}(\hat{y}_f) - \text{Dice}(\hat{y}_e) > 0$, we report the fraction of that gap recovered by the hybrid,

$$R_{\text{rec}}(k) = \frac{\text{Dice}(\hat{y}_k) - \text{Dice}(\hat{y}_e)}{\text{Dice}(\hat{y}_f) - \text{Dice}(\hat{y}_e)}, \tag{9}$$

which measures how much of the full decoder's benefit a $k\%$ spatial budget reproduces in output space, independent of runtime. Because the substitution is applied post hoc to precomputed outputs, R_{rec} is an opportunity bound rather than a deployable speedup: it isolates *how much benefit is spatially recoverable* from the separate systems question of *how cheaply* *the region can be produced*. When the full decoder holds no positive advantage ($\text{Dice}(\hat{y}_f) \le \text{Dice}(\hat{y}_e)$, as on datasets where the two decoders tie), the ratio is undefined and we instead report the absolute difference $\Delta\text{Dice}(k) = \text{Dice}(\hat{y}_k) - \text{Dice}(\hat{y}_e)$. Values $R_{\text{rec}}(k) > 1.0$ indicate over-recovery, where the selected region alone exceeds the full-volume gap, and are retained unclipped.

3.4.6. Factorial Main Effects and Interaction Analysis

The end-to-end pairs of Section 3.1 confound the two decoder factors with each other and with encoder weights, so they cannot say whether the observed behavior is driven by channel width, by high-resolution stages, or by their interaction. The frozen shared-encoder 2×2 grid removes this confound: the encoder is held fixed while channel capacity $C \in \{C_{384}, C_{48}\}$ and resolution-stage retention $RF \in \{RF_1, RF_2\}$ are varied independently. For any outcome metric M, we decompose the four-cell grid into two main effects and one interaction. The channel main effect averages the change from reducing channels over both resolution settings,

$$\begin{aligned}\Delta_C = \frac{1}{2}(&M(C_{48}, RF_1) + M(C_{48}, RF_2) \\ &- M(C_{384}, RF_1) - M(C_{384}, RF_2)),\end{aligned} \tag{10}$$

and the resolution-stage main effect symmetrically averages the change from removing high-resolution stages over both channel settings,

$$\begin{aligned}\Delta_{RF} = \frac{1}{2}(&M(C_{384}, RF_2) + M(C_{48}, RF_2) \\ &- M(C_{384}, RF_1) - M(C_{48}, RF_1)).\end{aligned} \tag{11}$$

Averaging each factor's effect over both levels of the other yields balanced, marginal estimates that are not biased by the specific setting of the nuisance factor. The interaction term measures how far the combined intervention departs from the sum of the two isolated effects,

$$\begin{aligned}\Delta_{C\times RF} =& M(C_{48}, RF_2) - M(C_{48}, RF_1) \\ &- M(C_{384}, RF_2) + M(C_{384}, RF_1).\end{aligned} \tag{12}$$

A near-zero $\Delta_{C\times RF}$ indicates the two factors act approximately additively — channel and resolution reductions can be reasoned about independently — whereas a large-magnitude interaction would signal that the factors must be co-designed. Applying this decomposition to the correctness-change metrics, rather than to Dice alone, attributes *where in space* each factor exerts its effect, linking the factorial back to the spatial axes above.

4. Experimental Setup

This section describes the target 3D backbones, clinical datasets, evaluation axes, and implementation setup.

4.1. Target Architectures and Intervention Specifications

We evaluate matched decoder pairs across three representative 3D medical image segmentation backbones: a large-kernel CNN (3D UX-Net [**?**]), a vision transformer (SwinUNETR [4]), and a ConvNeXt-style network (MedNeXt-M-K3 [5]). Interventions occur strictly within the decoder pathway (upsampling, skip fusion, and refinement) while keeping feature encoder architectures untouched. Table 2 details the intervention specifications across backbones.

Architectural and FLOP Profiling Protocol.. All computational complexity metrics (FLOPs/MACs) were measured using the fvcore profiling utility on a single 3D patch input tensor of shape $1 \times C \times 96 \times 96 \times 96$. To prevent cross-table confusion, we explicitly distinguish two specific 3D UX-Net full configurations evaluated in this study:

1. **End-to-End Reference Configuration** (Table 3): Trained end-to-end with default MONAI training pipelines containing auxiliary deep-supervision heads (53.01M params / 578.74 GMac).
2. **Frozen-Encoder Factorial Configuration** (Table 4): Evaluates the standalone decoder pathway with auxiliary heads detached under fixed unpadded patch profiling ($96 \times 96 \times 96$, 46.72M params / 657.76 GMac).

4.2. Factorial Configurations

To cleanly isolate the architectural mechanisms driving decoder activity, we perform a 2×2 factorial analysis on the 3D UX-Net backbone. By freezing the encoder and separating the decoding capacity along two structural axes (feature resolution and channel width), we evaluate whether performance and error variations originate from high-resolution spatial routing or abstract semantic capacity. Note that the factorial full configuration uses a separately instantiated decoder pathway to isolate these specific variables, and its FLOPs (657.76 GMac) are therefore not directly comparable with the tightly integrated end-to-end full configuration (578.74 GMac).
End-to-End Reference Full (Table 2): The baseline symmetric architecture.
Factorial-Analysis Full (Table 4): The isolated decoder pathway with all factorial features intact.

4.3. Clinical Datasets and Target Spacings

Experiments are conducted across three diverse 3D volumetric medical imaging datasets covering different anatomical regions and imaging modalities. Volumes are resampled to target spacings to match physical block side lengths of approximately 24 mm:

1. **BTCV Multi-Organ CT Dataset** [23]: 30 abdominal CT scans (resampled to $1.5\times1.5\times2.0$ mm^3; block dimensions $16\times16\times12$ voxels) with manual annotations for 13 organ structures.
2. **MSD01 BrainTumour MRI Dataset** [24, 25]: 484 multimodal 3D MRI volumes (resampled to $1.0\times1.0\times1.0$ mm^3; block dimensions $24 \times 24 \times 24$ voxels) with three foreground tumor classes.

Table 2: Backbone decoder specifications and efficient intervention parameters.

Architecture	Base Feature Size	Full Decoder (E_0)	Efficient Intervention (E_1)
3D UX-Net [?]	$C = 48$	C_{384}, 1× resolution	384 → 48 (87.5% cut), omit Stage 1
SwinUNETR [4]	$C = 48$	C_{384}, 1× resolution	384 → 48 (87.5% cut), omit Stage 1
MedNeXt-M-K3 [5]	$C = 32$	C_{256}, 1× resolution	256 → 32 (87.5% cut), omit Stage 1

3. **MSD08 HepaticVessel CT Dataset** [25]: 303 training volumes (resampled to $0.8 \times 0.8 \times 1.5$ mm^3; block dimensions $30 \times 30 \times 16$ voxels) focused on vascular and neoplastic structures.

4.4. Experimental Evaluation Axes

We evaluate decoder contribution along three complementary experimental axes rather than a full architecture-by-dataset Cartesian grid:

1. **Cross-Architecture Consistency Axis**: Evaluates matched full (E_0) vs. efficient (E_1) decoders across three backbones (3D UX-Net, SwinUNETR, MedNeXt-M-K3) on the canonical BTCV multi-organ CT dataset to test architectural consistency.
2. **Cross-Dataset Consistency Axis**: Evaluates matched full (E_0) vs. efficient (E_1) decoders using a fixed canonical backbone (3D UX-Net) across three diverse datasets (BTCV CT, MSD01 BrainTumour MRI, and MSD08 HepaticVessel CT) to test modality and anatomical generalization.
3. **Decoder-Isolated Factor Attribution Axis**: Evaluates a 2×2 factorial intervention design (E_0, Channel-only, Stage-only, E_1) using a frozen shared 3D UX-Net encoder on BTCV CT. Holding encoder feature representations strictly constant ensures that prediction shifts are primarily attributable to specific decoder interventions (channel width vs. resolution scaling).

4.5. Training Protocols and Implementation Setup

All models were implemented in PyTorch using an NVIDIA RTX 5090 GPU with CUDA 12.8. Models were trained for 45,000 optimizer iterations using bfloat16 mixed precision and the AdamW optimizer (initial learning rate 1×10^{-3} with cosine annealing, weight decay 1×10^{-5}) under a compound Dice-CrossEntropy loss function, evaluating every 250 iterations to match the EffiDec3D protocol. Sub-volume patches of $96 \times 96 \times 96$ voxels were extracted with an optimizer batch size of 1 volume per GPU, drawing 4 spatial crops per volume (yielding an effective batch size of 4 crops per optimization step). Standard 3D online data augmentations (random rotations, flips, intensity scaling) and dataset-specific intensity normalization were applied. Evaluation networks dynamically allocated a foreground sliding-window overlap of 0.7.

4.6. Data Splits and Model Selection

Every matched pair was trained and evaluated on identical subject-level training, validation, and test partitions, and partitioning was performed at the level of whole volumes so that no subject's sub-volumes leaked across splits. This shared-partition constraint is not a convenience but a requirement of the characterization: because our analysis is a *paired* voxel-level comparison between the full and efficient decoders, any discrepancy in the underlying data split would confound decoder-attributed correctness changes with data-attributed variance. Holding the split fixed ensures that every flip we measure reflects a difference in decoder computation on the same input, not a difference in which subjects each model saw.

Model checkpoints were selected exclusively by validation macro-Dice rather than training loss, which avoids selecting an overfit iterate, and all subsequent characterization analyses were computed strictly on held-out test subjects that were never used for training or checkpoint selection. Confidence intervals were estimated by subject-level paired bootstrap resampling (1000 resamples) over the held-out test set: the same subjects are resampled jointly for both decoders, preserving the within-subject correlation that a paired comparison exploits, so the reported intervals reflect subject-level variability rather than the artificially narrow bounds that voxel-count pooling would produce.

5. Results

In this section, we present observational characterization findings across our three experimental evaluation axes.

5.1. Accuracy–Computation Trade-off on BTCV

Table 3 summarizes aggregate computational complexity and segmentation performance under standard end-to-end training across three 3D backbones on BTCV CT. For each backbone, we evaluate the standard Full decoder (E_0) against an Efficient baseline (E_1) implementing the parameter reduction strategy of EffiDec3D [6]. The macroscopic picture is consistent across all three architectures: E_1 removes the large majority of decoder cost—up to a 91.4% FLOP reduction and a 5× latency reduction for 3D UX-Net—while the Mean Dice penalty stays within 1.45 points (79.18 → 77.73 for 3D UX-Net, 80.48 → 79.49 for SwinUNETR, 82.61 → 81.35 for MedNeXt-M-K3), and boundary error (HD95) even improves for two of the three backbones.

Read at face value, this table is the standard justification for decoder pruning: the extra computation looks nearly free to remove. Yet a Mean Dice difference of one point is a single scalar averaged over millions of voxels, and it says nothing about how many predictions the full decoder actually changed, in which direction, or where. The remainder of this section opens up that scalar. We show that behind these near-identical Dice scores the decoder performs a large amount of spatially structured correctness reorganization whose net effect is statistically indistinguishable from zero—the central observation that the aggregate view conceals.

5.2. Correctness Activity and Net Gain across Architectures

Evaluating voxel-level prediction shifts between matched E_0 and E_1 decoders reveals substantial gross correctness activity $R_{\text{act}} = R_+ + R_-$, whereas the net directional gain $R_{\text{net}} = R_+ - R_-$ remains substantially smaller due to bidirectional cancellation (Figure 3).

Subject-mean All-Volume net gain rates $R^{\text{all}}_{\text{net}}$ across models are: +0.046% (95% CI: [−0.020%, +0.110%]) for 3D UX-Net, −0.001% (95% CI: [−0.045%, +0.040%]) for SwinUNETR, and +0.052% (95% CI: [−0.010%, +0.108%]) for MedNeXt-M-K3. Evaluated over the Foreground-Union domain Ω_{fg}, net gain rates are $R^{\text{fg}}_{\text{net}} = +0.381\%$ for 3D UX-Net, −0.008% for SwinUNETR, and +0.446% for MedNeXt-M-K3. Across the three evaluated model pairs, subject-bootstrap confidence intervals for global net gain included zero, indicating no statistically detectable directional shift. In all models, positive voxel corrections $P(v)$ are largely offset by negative degradations $N(v)$.

5.3. Spatial Heterogeneity

The near-zero net gain of Section 5.2 is not the signature of a decoder that does little; it is the signature of intense but spatially confined activity. Two complementary stratifications—by physical distance to the anatomical boundary and by organ—localize where that activity lives, and both point to the same place.

Stratifying correctness activity $A_{\text{corr}}(v)$ by physical Euclidean distance $d_{GT}(v)$ to the true ground-truth boundary (Figure 4) reveals a sharp peak within the innermost 0–2 mm band and a rapid outward decay. Both positive corrections and negative degradations fall steeply across the 2–4 and 4–8 mm bands and are almost absent in homogeneous organ interiors ($d_{GT} > 8$ mm). The extra decoder computation therefore acts almost entirely in the thin transition zone around structure surfaces—exactly where partial-volume mixing makes voxel labeling ambiguous—while leaving confidently labeled interiors untouched. The pattern holds for all three backbones.

The organ-level breakdown (Figure 5) tells the same story along an anatomical axis. Small, morphologically complex structures with a high surface-to-volume ratio exhibit the largest correctness activity: the left and right adrenal glands reach positive-correction rates of 16.2%/16.9% in 3D UX-Net, 14.4%/14.8% in SwinUNETR, and 12.3%/14.0% in MedNeXt-M-K3. Large, compact structures such as the liver remain nearly inert (<3% activity), because their volume is dominated by interior rather than surface. Boundary proximity and small-complex-organ membership are thus two views of one geometric fact: decoder computation is spent where the foreground is mostly surface and saved where it is mostly interior.

5.4. Frozen-Encoder Factor Attribution

Table 4 reports our 2×2 factorial configuration grid on 3D UX-Net with frozen shared encoders on BTCV CT. Table 5 presents the main effects and interaction analysis across key outcome metrics:

- **High-Resolution Stage Removal** ($RF_1 \to RF_2$): Removing the high-resolution stage accounts for 657.8 GMac → 319.1 GMac FLOP savings. Retaining this high-resolution stage yields a net positive correction rate of +0.033% in the 0–2 mm physical GT boundary band and +0.031% in the 2–4 mm band, decaying to negative net values (−0.011%) in organ interiors (> 8 mm).

- **Channel Width Reduction** ($C_{384} \to C_{48}$): Reducing decoder channels compresses parameter count by 95% (46.7M → 2.2M) while exerting a spatially diffuse and net-neutral effect ($R^{\text{all}}_{\text{net}} = +0.027\%$, 95% CI: [−0.027%, +0.083%]).

5.5. Gain Concentration and Predictability

Figure 6 presents oracle benefit concentration curves. Across the three BTCV backbone pairs, the top 5% of spatial blocks captured at least 80% of positive gain ($K_{80} \leq 5\%$), and the top 10% captured up to 99.98%.

We evaluate whether cheap inference-time signals can locate regions where decoder computation alters correctness (Location AUROC) and whether they can further discriminate the direction of those changes (Direction AUROC) (Table 6 and Figure 7):

- **Activity Location AUROC**: Predictive entropy reliably identifies active blocks ($A_{\text{corr}}(r) > 0$), achieving high location AUROCs of 0.900 for 3D UX-Net (AUPRC 0.371), 0.913 for SwinUNETR (AUPRC 0.351), and 0.907 for MedNeXt-M-K3 (AUPRC 0.389).

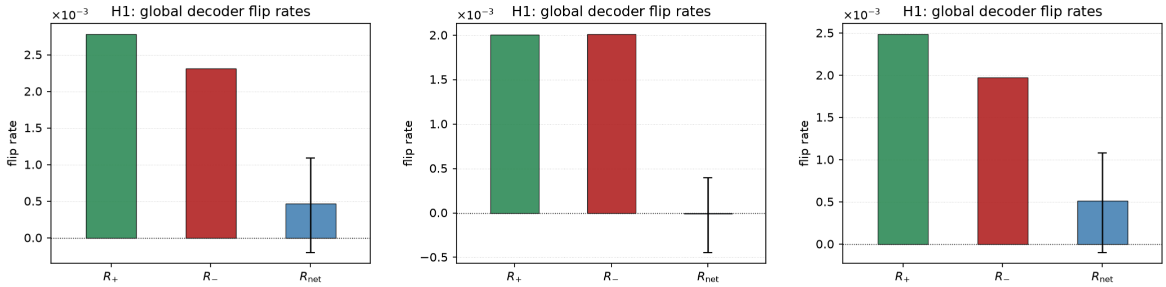
- **Directional Utility Discrimination**: Predictive entropy provides strong activity localization but only weak directional discrimination ($G(r) > 0$ vs. $G(r) < 0$), yielding direction AUROCs modestly above chance: 0.591 for 3D UX-Net, 0.560 for SwinUNETR, and 0.556 for MedNeXt-M-K3.

- **TTA Mutual Information (TTA-MI)**: Multi-sample epistemic disagreement $I(v)$ over $K = 8$ spatial view transformations yields modest direction AUROCs of 0.539 for 3D UX-Net and 0.570 for SwinUNETR.

- **Conditional Density Overlap**: Figure 7 visualizes conditional entropy density distributions $p(H \mid P(v) = 1)$ and $p(H \mid N(v) = 1)$, showing near-identical overlap.

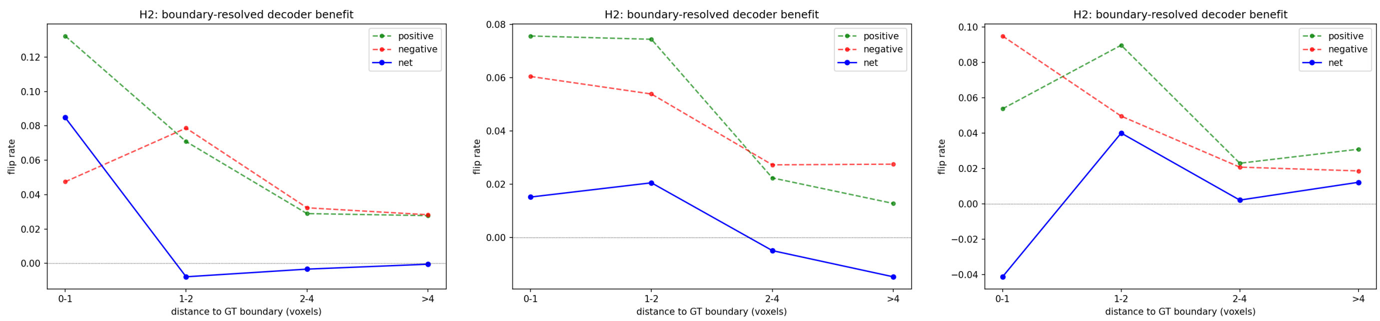
Table 3: Quantitative comparison of Full (E_0) vs. Efficient (E_1) decoders on the BTCV 13-organ dataset across three backbones (Cross-Architecture Consistency Axis).

Architecture	Variant	Params (M) ↓	FLOPs (GMac) ↓	Latency (ms) ↓	Mean Dice (%) ↑	Mean HD95 (mm) ↓
3D UX-Net	Full (E_0)	53.01	578.74	40.6	79.18	9.04
	Effi (E_1)	3.16	49.49	8.0	77.73	11.82
SwinUNETR	Full (E_0)	62.19	308.24	37.0	80.48	8.88
	Effi (E_1)	11.21	55.84	16.7	79.49	8.90
MedNeXt-M-K3	Full (E_0)	17.56	106.34	27.8	82.61	7.25
	Efficient (E_1)	5.80	49.54	12.9	81.35	6.77



(a) 3D UX-Net Global Flips (b) SwinUNETR Global Flips (c) MedNeXt-M-K3 Global Flips

Figure 3: Voxel-level net gain rates across 3D UX-Net, SwinUNETR, and MedNeXt-M-K3 backbones under Full (E_0) vs. Efficient (E_1) decoders.



(a) 3D UX-Net Boundary Flips (b) SwinUNETR Boundary Flips (c) MedNeXt-M-K3 Boundary Flips

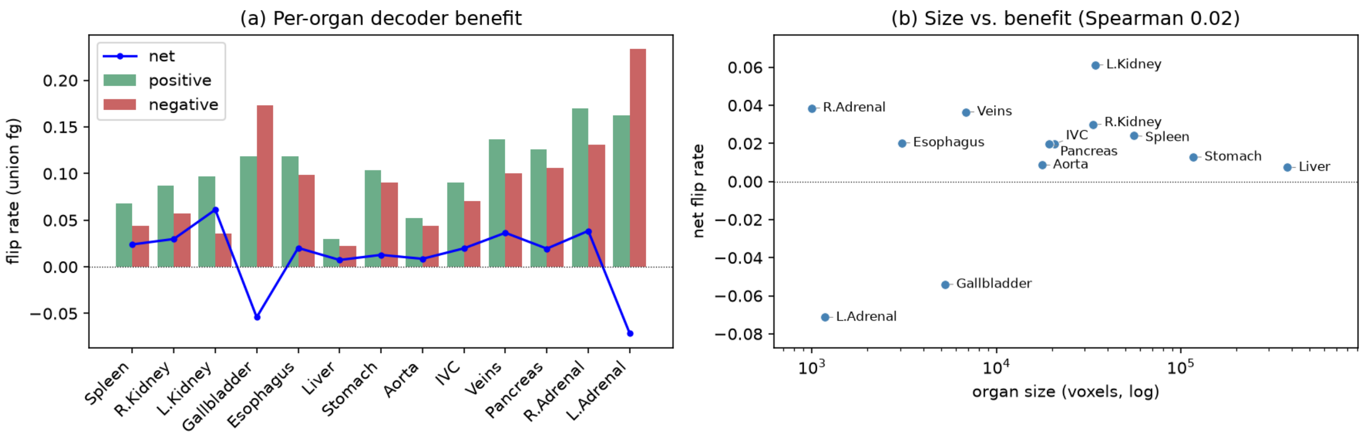
Figure 4: Spatial ground-truth boundary concentration of prediction flips across the three evaluated backbones.

Table 4: 2×2 Factorial analysis of decoder channel reduction (C_{dec}) and resolution scaling (RF) on 3D UX-Net and BTCV CT (Decoder-Isolated Factor Attribution Axis).

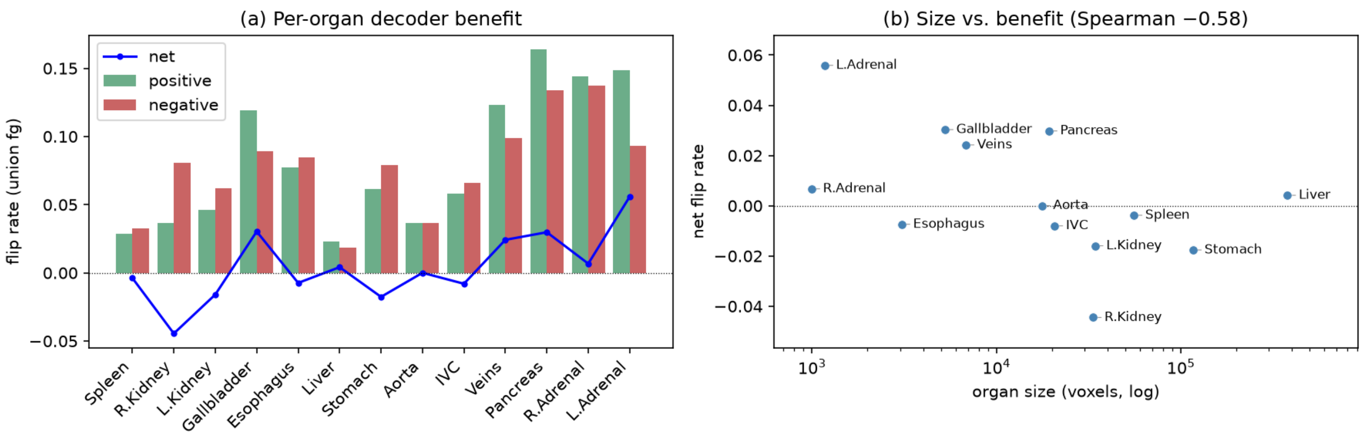
Configuration	C_{dec}	RF	Params (M)	FLOPs (GMac)	Latency (ms)	Mean Dice (%)
Full (C_{384}, RF_1)	384	1	46.72	657.76	43.3	80.44
Chan-Only (C_{48}, RF_1)	48	1	2.24	388.15	35.2	79.80
Res-Only (C_{384}, RF_2)	384	2	46.32	319.10	16.1	77.97
Effi (C_{48}, RF_2)	48	2	1.84	49.49	8.0	78.35

Signal-guided retrospective recovery.. Although entropy cannot tell a correction from a degradation, it locates activity well enough that concentrating full-decoder computation on the highest-entropy blocks recovers the macro-Dice gap. Figure 8 presents retrospective macro-Dice trajectories under output hybridization using various predictive signals:

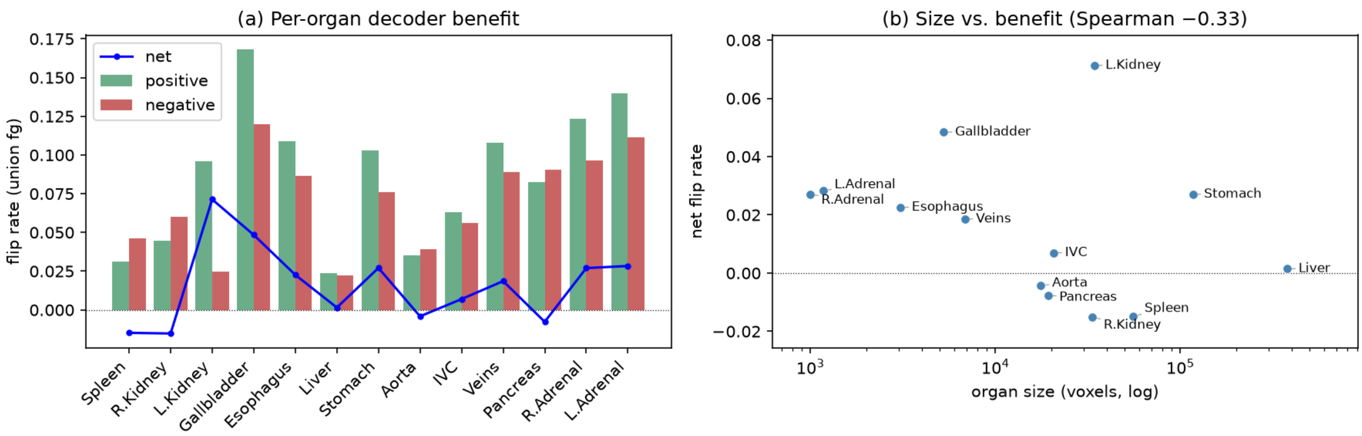
- At a **5% block budget**, entropy-guided output substitution recovers **60% of the full decoder's Dice gap** over E_1 for 3D UX-Net (vs. 8% for random selection).
- At a **20% block budget**, entropy-guided output substitution achieves **full recovery of the macro-Dice gap** across all three backbones—3D UX-Net, SwinUNETR, and MedNeXt-M-K3 (entropy recovery ≥ 100%, vs. 17–20% for random selection).



(a) 3D UX-Net Organ-Level Flips



(b) SwinUNETR Organ-Level Flips



(c) MedNeXt-M-K3 Organ-Level Flips

Figure 5: Organ-level flip breakdown across backbones on the BTCV CT multi-organ dataset under Full (E_0) vs. Efficient (E_1) decoders.

5.6. Cross-Dataset Replication

To evaluate whether the observations discovered on BTCV generalize across imaging modalities and anatomical targets, Table 7 summarizes these metrics across all three datasets for the 3D UX-Net reference architecture. Across datasets, gross bidirectional correctness activity, positive-gain concentration (e.g., the strong concentration pattern persisted on MSD08 with $K_{80} \leq 5\%$), and the activity–direction predictability gap remained observable. However, the magnitude and sign of net gain were task-dependent: on MSD08, the efficient decoder outperformed the full decoder, producing a negative foreground-union net gain.

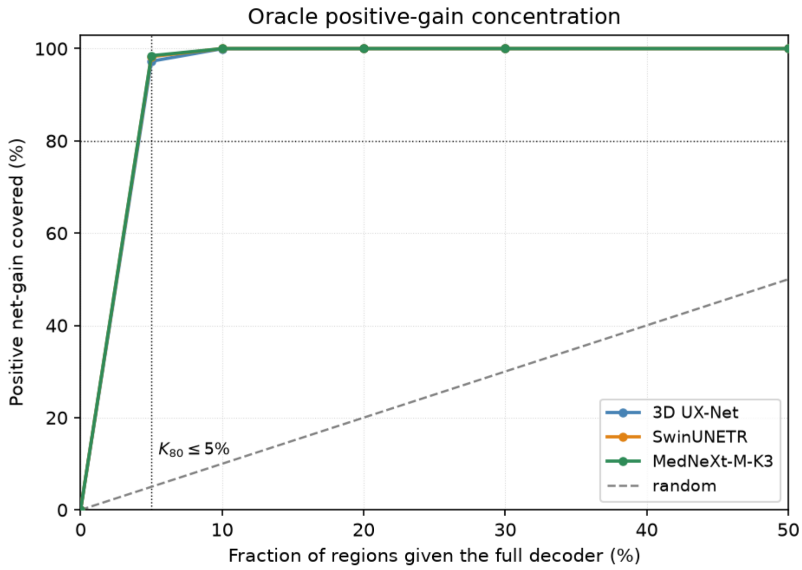
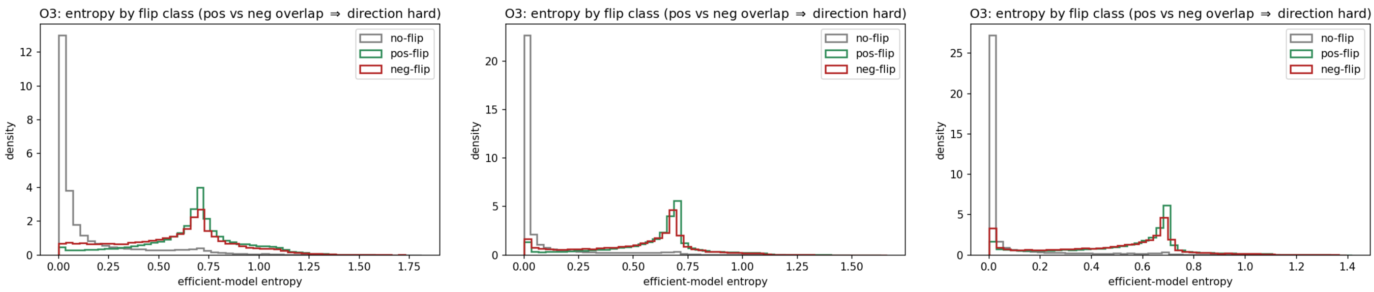
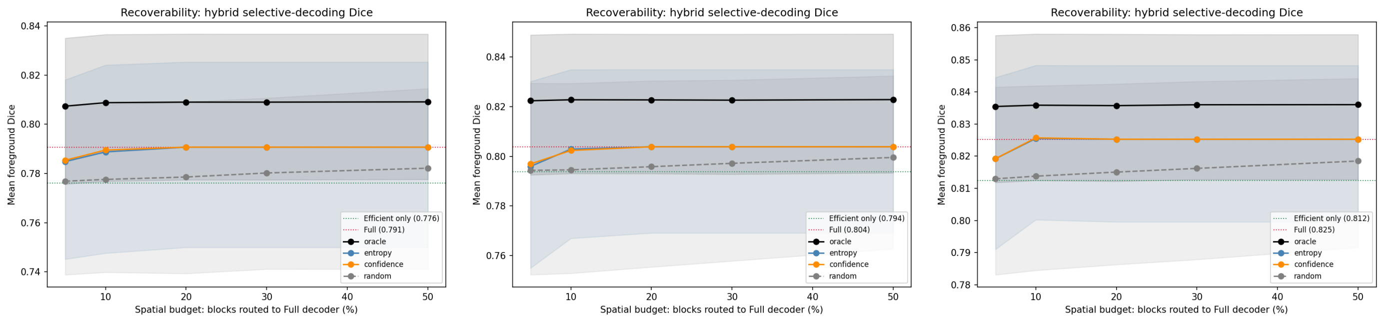


Figure 6: Oracle positive-gain concentration for the three backbones on a single color-coded axis. The gray dashed line is the random-selection baseline. In all three architectures the positive net gain is heavily concentrated, satisfying $K_{80} \leq 5\%$: the top 5% of spatial blocks already cover $\geq 97\%$ of the total positive gain, and the top 10% cover essentially all of it.



(a) 3D UX-Net Flip Entropy Density (b) SwinUNETR Flip Entropy Density (c) MedNeXt-M-K3 Flip Entropy Density

Figure 7: Conditional entropy density distributions for positive corrections (P) and negative degradations (N).



(a) 3D UX-Net Macro Recovery (b) SwinUNETR Macro Recovery (c) MedNeXt-M-K3 Macro Recovery

Figure 8: Signal-guided retrospective macro-Dice trajectories under output hybridization at increasing spatial budgets.

6. Discussion

This study shifts the evaluation of decoder efficiency from global FLOPs reduction to spatial computation dynamics. This section explains the underlying mechanisms, addresses the metric weighting paradox, outlines architectural implications, and discusses limitations.

Table 5: Factor main effects (Δ_C, Δ_{RF}) and interaction ($\Delta_{C\times RF}$) across metrics on 3D UX-Net (BTCV CT).

Outcome Metric	Channel Δ_C	Stage Δ_{RF}	Interaction $\Delta_{C\times RF}$
Mean Dice (%)	−0.13	−1.96	+1.02
Params (M)	−44.48	−0.40	+0.00
FLOPs (GMac)	−269.61	−338.66	+0.00
Boundary Net Gain (%)	−0.020	+0.033	−0.004

Table 6: Inference-time predictability of block-level correctness activity (Location AUROC) and directional utility (Direction AUROC) across evaluated backbones.

Backbone	Location AUROC	Direction AUROC
3D UX-Net	0.900	0.591
SwinUNETR	0.913	0.560
MedNeXt-M-K3	0.907	0.556

6.1. Decoder Activity versus Net Benefit

Our findings show that additional decoder capacity does not simply act as a monotonic accuracy enhancer. Instead, it generates substantial gross correctness activity $A_{\text{corr}}(v) = P(v) + N(v)$, which is largely offset by bidirectional cancellation ($P(v) \approx N(v)$). Positive corrections and negative degradations coexist within ambiguous boundary regions, producing substantial local activity but much smaller aggregate net gain. Rather than systematically fixing errors volume-wide, high-capacity decoders reallocate predictions near ambiguous boundaries.

6.2. Coexistence of Small Voxel Net Gain and Macro Dice Improvement

A key question arises: *How can additional decoder computation improve macroscopic Dice on BTCV and MSD01 while producing a small all-volume net voxel gain, and conversely, decrease macroscopic Dice on MSD08?*

This coexistence stems from the fundamental difference in metric weighting:

1. **Voxel Accuracy Weighting**: Global voxel correctness is heavily dominated by vast background regions and large organ interiors, where predictions are stable under both full and efficient decoders.
2. **Macro Dice Sensitivity**: The macro-averaged Mean Dice assigns equal weight to anatomical classes and is foreground-sensitive. Positive voxel corrections $P(v)$ occurring at thin anatomical boundaries or small, morphologically volatile organs (such as the adrenal glands) exert a disproportionately large impact on macroscopic Dice scores.

Together, these results show that aggregate decoder utility is task-dependent, while all-volume voxel shifts remain small relative to foreground-localized changes.

6.3. Implications for Adaptive Decoder Design

These findings do not directly prescribe a specific adaptive decoder architecture. Instead, they suggest three practical constraints and directions for future utility-aware decoder design:

1. **Separate activity localization from utility estimation**: Predictive uncertainty is effective for locating regions where decoder computation may alter correctness, but it is insufficient for determining whether those alterations will be beneficial. Future adaptive decoders should therefore separate candidate-region proposal from expected-gain estimation.
2. **Decouple globally shared decoding from local high-resolution refinement**: The frozen-encoder factorial analysis suggests that high-resolution decoding stages contribute most strongly near anatomical boundaries, whereas channel-capacity changes produce more spatially diffuse effects. This motivates globally lightweight decoders with selectively activated high-resolution refinement modules.
3. **Treat additional computation as a task-dependent intervention**: Because the sign and magnitude of net gain vary across datasets—and the efficient decoder outperforms the full decoder on MSD08—additional computation should not be assumed to be monotonically beneficial. Routing policies should preserve the efficient prediction unless the expected gain of refinement exceeds its computational cost.

Together, these requirements establish an opportunity bound that motivates globally lightweight decoders that activate local refinement only when its expected benefit justifies its cost.

6.4. Clinical Stratification of Decoder Utility

The spatial heterogeneity we observe is not clinically uniform: it maps onto a stratification of anatomical structures whose segmentation errors carry very different clinical consequences. Reading our per-organ and boundary-resolved results through this lens yields three tiers with distinct implications for where decoder computation should be spent.

Tier 1 — large compact organs (e.g., liver, spleen, kidneys). These structures are volume-dominated rather than surface-dominated, and show negligible correctness activity ($< 3\%$) under decoder simplification. For clinical tasks defined over their bulk—organ volumetry, steatosis or lesion-load quantification, and organ-at-risk delineation for radiotherapy where the interior dominates the contour—the efficient decoder is effectively equivalent to the full one. Here the latency savings can be taken safely, which is directly relevant to time-constrained intraoperative and high-throughput screening settings.

Tier 2 — small, thin, or morphologically complex structures (e.g., adrenal glands, gallbladder, hepatic vessels). These exhibit the highest correctness activity (adrenal positive-correction rates up to 16.9%), and it is concentrated in the 0–2 mm boundary band. Clinically these are exactly the structures where a one- or two-voxel boundary shift changes a decision: tumor-margin definition for resection planning, vessel-wall delineation for catheter navigation and perfusion territory mapping, and small-gland localization for functional assessment. For these tasks the full decoder's boundary corrections are where its value lives, and a fixed efficient decoder should not be deployed without the boundary-targeted refinement (or entropy-guided routing) that recovers exactly this contribution.

Table 7: Cross-dataset generalization validation using 3D UX-Net across three clinical imaging datasets (Cross-Dataset Consistency Axis).

Dataset	**Full Dice** (%)	**Effi Dice** (%)	**Activity** $R_{\text{act}}^{\text{all}}$ (%)	**Net Gain** $R_{\text{net}}^{\text{all}}$ (%)	**Activity** $R_{\text{act}}^{\text{fg}}$ (%)	**Net Gain** $R_{\text{net}}^{\text{fg}}$ (%)	**Oracle** K_{80}	**Location AUROC**	**Direction AUROC**
BTCV CT	79.18	77.73	3.42	+0.046	12.84	+0.381	≤ 5%	0.900	0.591
MSD01 BrainTumour	75.55	74.95	0.46	+0.038	18.10	+2.86	≤ 5%	0.869	0.558
MSD08 HepaticVessel	57.51	58.63	0.20	−0.002	22.99	−4.89	≤ 5%	0.865	0.520

Tier 3 — tasks where more computation is not safer. On MSD08 HepaticVessel the efficient decoder slightly *outperformed* the full decoder, a reminder that added capacity is not monotonically beneficial and can inject boundary noise on already-thin structures. Clinically this argues against a reflexive "use the largest model" policy: for some vascular and lesion targets the conservative efficient prediction is preferable, and refinement should be gated on demonstrated expected gain rather than applied by default.

Crucially, the population-level net-neutrality we report (Section 5.2) means that at the level of aggregate, cohort-averaged accuracy the efficient decoder is clinically non-inferior in our evaluations; the stratification above governs *which structures and which tasks* nonetheless warrant the extra computation. This separation—safe global simplification, targeted local refinement—is the practical clinical takeaway of the characterization.

6.5. Limitations

This study has several limitations. First, the controlled interventions follow the channel-reduction and high-resolution stage-omission principles of EffiDec3D, and factor-level attribution is performed only on 3D UX-Net and BTCV. Although the end-to-end observations are evaluated across multiple architectures and datasets, the factorial effects may differ under other decoder designs or efficiency interventions.

Second, the end-to-end full and efficient models are trained independently under matched protocols. The frozen shared-encoder experiment controls variation in feature representations, but independent decoder initialization and optimization trajectories may still contribute to prediction differences.

Third, the routing analysis is based on retrospective output-space hybridization. It therefore provides an opportunity bound on selective decoder allocation rather than demonstrating a deployable dynamic execution system or proportional reductions in runtime latency and hardware-level computation.

Finally, predictability is evaluated primarily using predictive entropy and TTA mutual information on three public benchmarks. Broader clinical cohorts and learned or feature-based utility predictors may reveal additional information about directional decoder gain.

7. Conclusion

We systematically characterized decoder computation in 3D medical image segmentation, evaluating matched full and efficient decoder pairs across three backbones and three datasets. The characterization separates two quantities that aggregate metrics conflate—where the decoder changes correctness (activity) and whether it helps (net gain)—and shows they diverge: added decoder capacity is highly active yet nearly net-neutral, its residual benefit is concentrated in the top 5% of spatial blocks ($K_{80} \leq 5\%$) along a thin anatomical-boundary band, and inference-time uncertainty locates *where* correctness changes far better than it predicts *whether* they improve.

Beyond these specific findings, the characterization points to a design principle for the next generation of segmentation networks. The prevailing symmetric encoder–decoder allocates dense, high-resolution computation uniformly across the volume, yet we find that the bulk of this computation is spatially misallocated: it is spent where its net effect is indistinguishable from zero. This argues for decoders that are not monolithic dense pathways but two-part systems—a globally lightweight base decoder that produces the volume-level segmentation, coupled to a sparse, conditionally executed high-resolution refinement pathway that is invoked only on the small boundary fraction where it pays off. Realizing such a design hinges on a component the field does not yet have: a *utility* predictor that estimates expected directional gain, distinct from the uncertainty and location signals that today's adaptive methods rely on and that we show are insufficient on their own. Our concentration and predictability results quantify both the opportunity (a large fraction of decoder cost is removable) and the missing ingredient (directional utility estimation), giving concrete targets for that line of work. The encoder, which we deliberately hold fixed and which still accounts for the majority of inference cost, remains an open axis for the same style of analysis.

Clinically, the characterization translates into a stratified deployment guideline rather than a single verdict. Because the decoder's net effect over the volume is negligible, efficient decoders can be deployed with confidence in latency-critical settings such as intraoperative guidance and high-throughput screening, and for bulk tasks over large compact organs. The sparse computation that does matter is concentrated at the anatomical boundaries—tumor margins, vessel walls, and organ interfaces—that most directly shape surgical and radiotherapy decisions, and it is precisely there, and on small morphologically complex structures, that boundary-targeted refinement should be retained. Directing decoder capacity by this stratification, rather than uniformly, offers a principled path to models that are simultaneously efficient and clinically dependable where it counts.

CRediT authorship contribution statement

Author One: Conceptualization, Methodology, Writing – original draft. **Author Two**: Data curation, Software. **Author**

Three: Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used in this study (BTCV/Synapse, MSD01 BrainTumour, and MSD08 HepaticVessel) are publicly available.

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Supplementary Material: Qualitative Visualizations for 3D UX-Net and MedNeXt-M-K3

This section provides supplementary qualitative visualizations for 3D UX-Net and MedNeXt-M-K3 backbones, complementing the SwinUNETR qualitative motivation example in Figure 2 of the main manuscript.

Supplementary Material: Entropy Monotonicity with Block-Level Error Rate

Figure S3 visualizes the block-level spatial correlation between mean predictive entropy and block-level error rate across all three evaluated backbones. The monotonic relationship confirms that entropy reliably localizes active, error-prone regions (Location AUROC), even though its directional discrimination ability remains weak.

Supplementary Material: Additional Implementation Details

TTA-MI (Test-Time Augmentation Mutual Information) Setup

For evaluating the predictive power of multi-view epistemic disagreement against spatial entropy, we utilized 8-view Test-Time Augmentation (TTA). Each volume was forwarded through the network across all 8 spatial flip combinations (x, y, z axes). The Mutual Information (TTA-MI) was computed as the difference between the entropy of the mean prediction and the mean of the individual view entropies. Despite this computationally expensive multi-pass estimation, TTA-MI provided only marginally better directional discrimination (Direction AUROC 0.570 for SwinUNETR, 0.539 for 3D UX-Net) compared to single-pass predictive entropy, further reinforcing our conclusion that uncertainty measures are powerful activity indicators but weak directional utility classifiers.

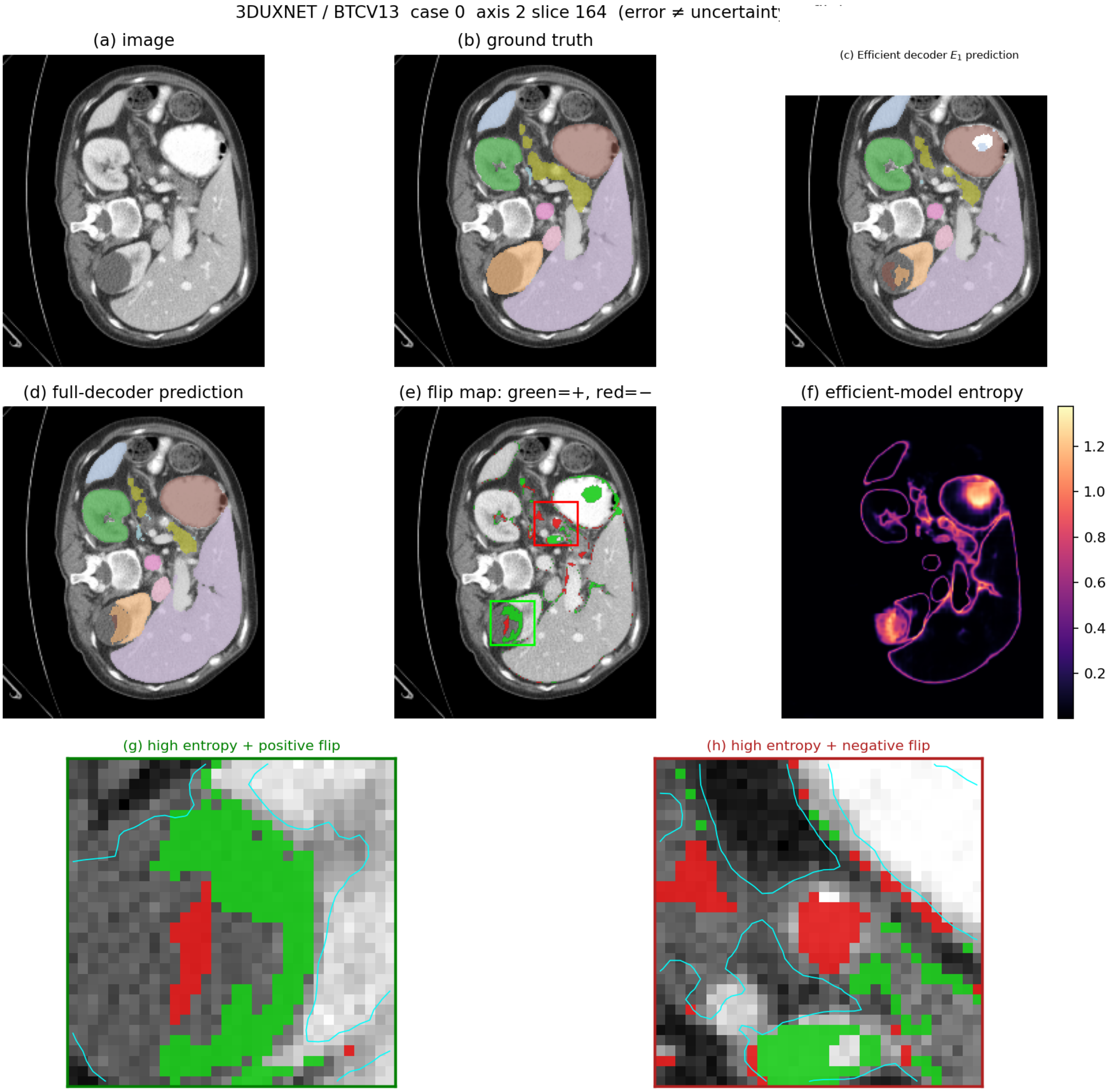


Figure S1: Supplementary Figure S1: Qualitative motivation visualization for 3D UX-Net on BTCV CT multi-organ dataset under Full (E_0) vs. Efficient (E_1) decoders.

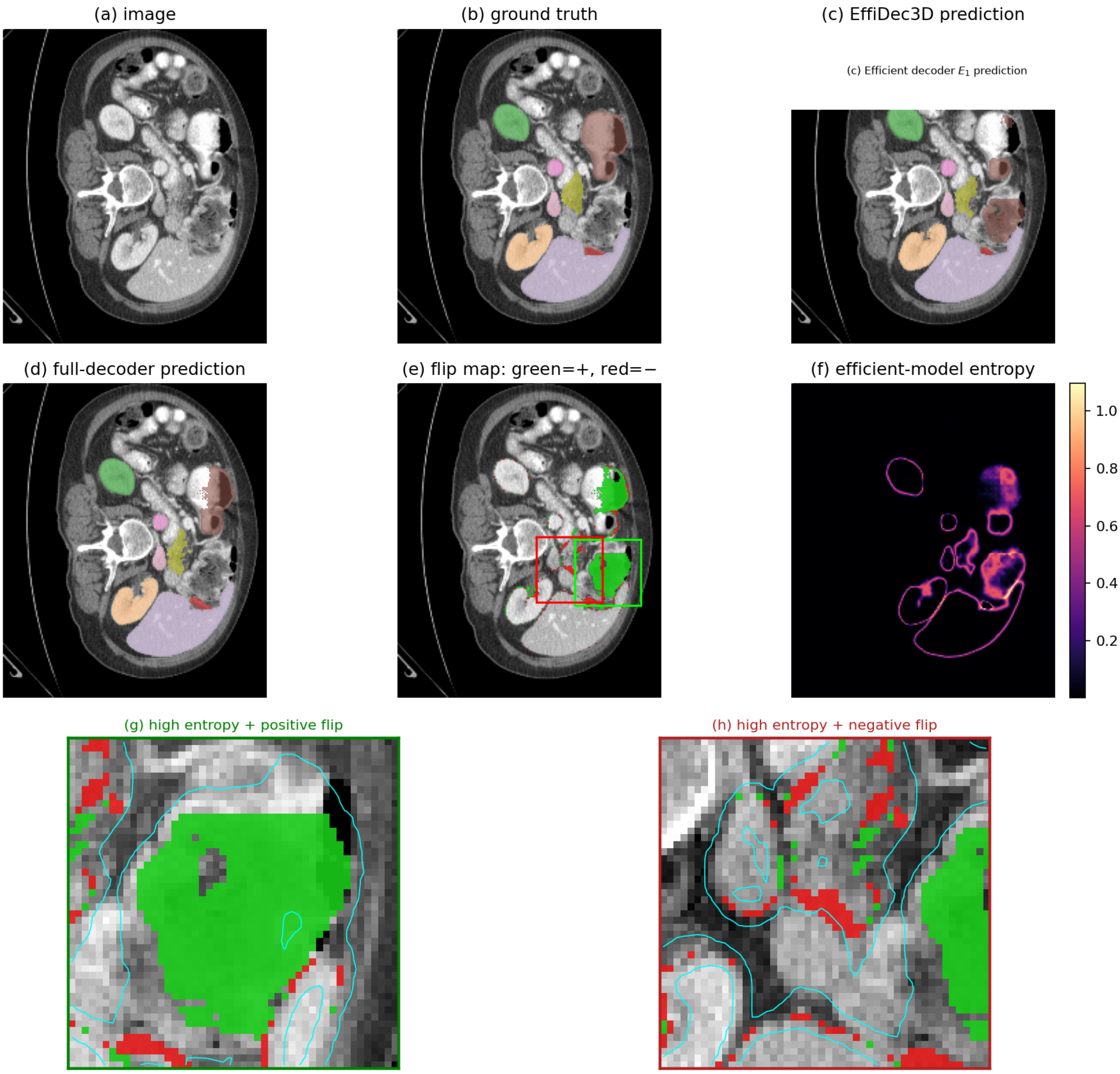
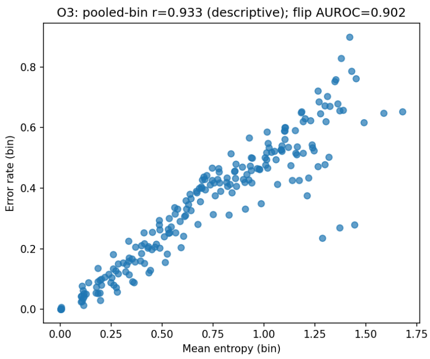
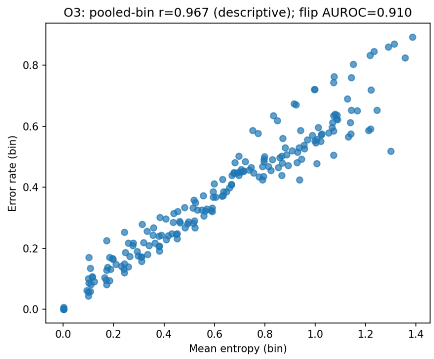


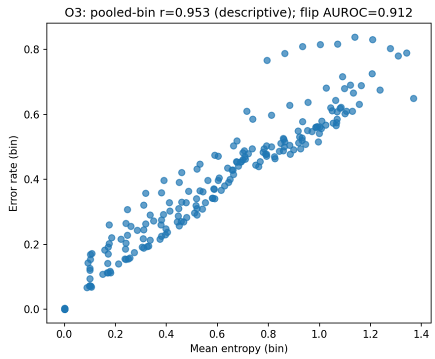
Figure S2: Supplementary Figure S2: Qualitative motivation visualization for MedNeXt-M-K3 on BTCV CT multi-organ dataset under Full (E_0) vs. Efficient (E_1) decoders.



(a) 3D UX-Net Error Correlation



(b) SwinUNETR Error Correlation



(c) MedNeXt-M-K3 Error Correlation

Figure S3: Supplementary Figure S3: Correlation between mean predictive entropy and block-level error rate across backbones.